Practical 6: Petrography of placer deposits

In this practical we will explore the petrography and formation of placer gold deposits. Through the evaluation of the gold potential in the Witwatersrand braided river system, using data from boreholes.

Activity 1 – Petrography

As a group, examine the slab of typical gold-bearing conglomerate (slab about 15cm high), using a binocular microscope.

Describe:

- the nature of the lower contact
- the lithology, grain size and degree of rounding of components
- can you identify any gold and/or pyrite grains?

What do your observations tell you about depositional processes in the conglomerate?

Activity 2 – Evaluation of placer gold deposit

Note: We will start this exercise in the practical, but you are encouraged to finish it in your own time after the practical. Suggested solutions will be posted next week on Blackboard.

Vaal Gold Ltd have recently taken a new exploration block adjacent to their massive Randfontein gold field, part of the Witwatersrand gold deposit. From Randfontein experience, the Vaal Gold geologists know there is potential for more gold in adjacent areas such as this new block. Extensive drilling has indeed encountered gold-bearing conglomerate reefs. The geologists have determined conglomerate thickness, pebble grain size, orientation of paleoflow directions, and gold concentrations in these drill cores. From this dataset, they can now evaluate the orientations of paleo-channels, and predict the best gold-bearing areas to evaluate whether mining is worthwhile.

Data set:

Figure 1 shows a map of the license block with locations of the exploration wells.

Table 1 contains the following data:

- True conglomerate thickness in cm
- Average pebble grain size (in phi – ϕ , which is an alternative scale to mm; ϕ values between ~ -2 to ~ -4 correspond to ~4-16 mm – see table in appendix)
- Gold concentration in ppm

Paleoflow directions (arrows on the map) were measured from sedimentary structures like cross bedding. We know from the lectures the conglomerate reefs represent channel deposits in a braided fluvial system.

Task:

Use the information from the map and from Table 1 to evaluate gold occurrence in this block.

1. Plot the **conglomerate thickness**, the **mean pebble size**, and the **gold concentration** for each sample on the map. Use one copy of the base map for each parameter, and label the map accordingly.
2. Make contour maps by drawing the isolines for the 3 parameters you plotted, i.e. the lines that connect all points of equal value. Use the following isolines: conglomerate thickness 40-80-120-160-200 cm; mean pebble size: every 0.5 phi; gold concentration: every 0.5 ppm. You can color in the area between the isolines.

Questions:

1. Can you identify the main paleo-channels? Draw them on any of the maps you created.
2. Identify the most prospective areas, i.e. highest grades. Do you think this block fulfills expectations?

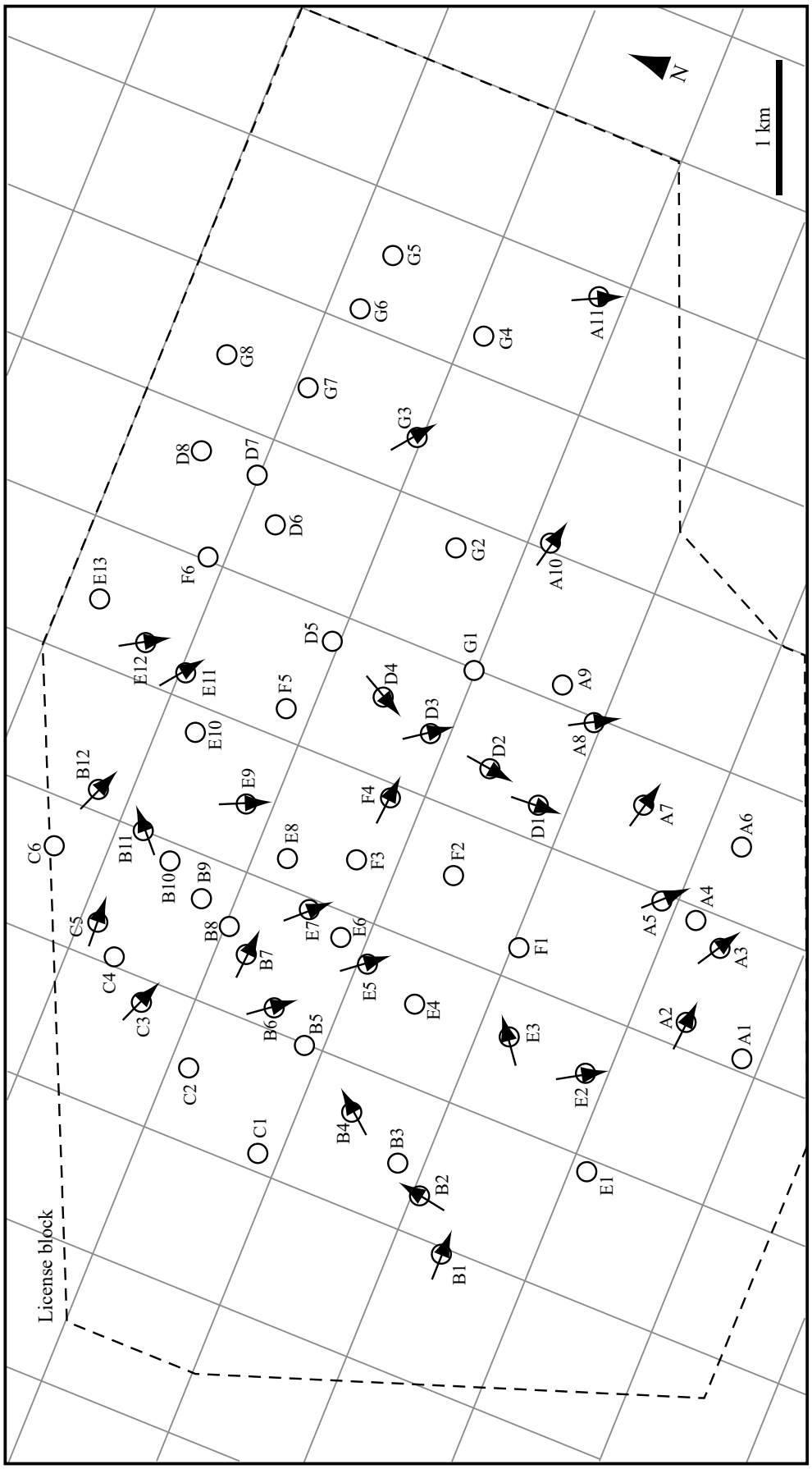


Figure 1: Location map

Borehole	Conglomerate thickness (cm)	Mean pebble size (phi)	Au concentration (ppm)
A1	35	-2.4	2.3
A2	70	-3.2	4.4
A3	60	-2.8	4.4
A4	35	-2.6	3.4
A5	65	-2.6	3.7
A6	25	-2.3	N/A
A7	95	-3.0	4.3
A8	110	-2.8	3.2
A9	N/A	-2.5	2.8
A10	35	-2.2	2.3
A11	105	-1.8	3.1
B1	60	-4.3	3.0
B2	65	-4.2	2.9
B3	45	-4.0	2.8
B4	40	-3.8	2.6
B5	35	-3.6	2.2
B6	45	-3.7	2.8
B7	80	-3.8	2.6
B8	85	-3.6	2.8
B9	95	-3.4	3.5
B10	210	-3.6	4.1
B11	205	-2.7	4.6
B12	40	-3.2	3.3
C1	20	-3.9	1.2
C2	35	-3.9	2.6
C3	90	-4.0	3.3
C4	65	-3.9	3.2
C5	40	-3.9	2.6
C6	25	-3.6	N/A
D1	65	-3.0	4.5
D2	80	-2.9	3.3
D3	165	-2.9	5.2
D4	210	-2.8	5.5
D5	75	-2.7	2.2
D6	125	-2.8	3.1
D7	75	-2.6	1.9
D8	40	-2.3	1.8
E1	30	-3.3	1.8
E2	60	-3.3	4.2
E3	90	-3.5	4.1
E4	80	-3.4	3.6
E5	35	-3.5	2.7
E6	45	-3.4	2.6
E7	85	-3.2	N/A
E8	N/A	N/A	2.6
E9	120	-3.1	2.8
E10	205	-3.2	3.4
E11	150	-3.1	3.2
E12	100	-2.9	2.8
E13	30	-2.8	2.2
F1	90	-3.3	3.7
F2	50	-3.2	3.1
F3	80	-3.2	3.0
F4	90	-3.1	4.1
F5	105	-2.9	2.7
F6	140	-2.9	3.7
G1	140	-2.7	4.8
G2	25	-2.5	1.6
G3	130	-2.6	2.8
G4	100	-2.0	3.1
G5	40	-2.2	2.0
G6	80	-2.5	3.3
G7	60	-2.2	2.5
G8	45	-2.3	2.3

Table 1

Length (mm)			Class	Sediment/ rock name
	ϕ		block	mega-conglomerate
4096	−12			
2048	−11	vc	boulder	gravel conglomerate
1024	−10	c		
512	−9	m		
256	−8	f		
128	−7	c	cobble	
64	−6	f		
32	−5	vc	pebble	
16	−4	c		
8	−3	m		
4	−2	f		
2	−1		granule	
1	0	vc	sand	sand sandstone
0.50	1	c		
0.25	2	m		
0.125	3	f		
0.063	4	vf		
0.031	5	c	silt	silt siltstone
0.015	6	m		
0.008	7	f		
0.004	8	vf		
			clay	clay claystone

Grain size chart to compare mm and phi values